VPA IB DP CS — Year 2 Semester 1 Assessment Pack

Year 2 Semester 1 Assessment Pack 评估包 Pínggū Bāo

HL Topics: 5–6 | SL Focus: Revision + IA

1. HL Assessment Calendar

Week	Assessment	Type	Marks
5	Topic 5 Progress Check	Formative	30
9	Topic 5 End-of-Topic Test	Summative	50
14	Topic 6 Progress Check	Formative	25
17	Topic 6 End-of-Topic Test	Summative	50
18	Semester 1 Examination	Summative	100 (HL Paper 1)

2. SL Assessment Calendar

Week	Assessment	Type	Marks
1	Diagnostic Test (Topics 1–4)	Baseline	40
3	Topic 1–2 Mock	Formative	30
6	Topic 3–4 Mock	Formative	40
9	SL Revision Test 2	Summative	50
16	Full Paper 1 + Paper 2 Mock	Summative	130
17	IA Final Submission	Summative	30% of total

3. HL Sample Questions (Topics 5–6)

Q1 — Define [2]

(a) Define the term pointer. (指针 zhǐzhēn) [1]

(b) Define the term deadlock. (死锁 sǐsuǒ) [1]

Q2 — Describe [4]

Describe the process of inserting a new node at the start of a singly linked list.

Q3 — Construct [5]

Given the following sequence of numbers: 50, 30, 70, 20, 40, 60, 80, construct a binary search tree showing the final structure.

Q4 — Trace [6]

Trace Dijkstra's algorithm on the following weighted graph, starting from vertex A. Show the shortest distance to each vertex at each step.

Q5 — Explain [6]

Explain why paging can cause internal fragmentation while segmentation can cause external fragmentation.

Q6 — Evaluate [10]

Evaluate the use of round robin scheduling in a multi-user operating system where some users run CPU-intensive batch jobs and others run interactive applications.

Q7 — Discuss [12]

Discuss the ethical implications of relying on autonomous control systems in safety-critical applications such as medical devices or self-driving vehicles.

4. SL Revision Sample Questions

Q8 — Explain [5]

Explain the purpose of cache memory in improving CPU performance.

Q9 — Compare [6]

Compare packet switching and circuit switching for transmitting real-time video data.

Q10 — Construct [6]

A shop wants to store information about products (ProductID, Name, Price, StockLevel) and suppliers (SupplierID, Name, Phone). A product is supplied by exactly one supplier. A supplier can supply many products.

Construct an ER diagram for this system. [4]

Construct a flowchart for a process that checks if stock level is below 10 and outputs "Reorder" if true. [2]

Q11 — Trace [4]

Trace the following algorithm for the input [5, 2, 8, 1, 9], showing the array after each pass:

for i ← 0 to 3 do

for j ← 0 to 3□i do

if A[j] > A[j+1] then

temp ← A[j]

A[j] ← A[j+1]

A[j+1] ← temp

end if

end for

end for

5. IA Assessment Criteria Reminder

Criterion	Max Marks	Key Requirements
A. Planning	6	Real client, real problem, consultation evidence, rationale, success criteria
B. Design	6	Record of tasks, system overview, algorithm design, test plan
C. Development	12	Complex techniques, code with commentary, video evidence
D. Evaluation	4	Testing against success criteria, client feedback, future improvements

6. Examination Technique Checklist

Before the exam:

☐ Know the command terms and what they require

☐ Memorise key definitions (bilingual if needed)

☐ Practise binary calculations to fluency

☐ Can draw ER diagrams, flowcharts, logic circuits from memory

☐ Know sorting/searching algorithms and their complexities

During the exam:

☐ Read the question carefully — highlight command term

☐ Plan 12-mark essays before writing (2–3 minutes)

☐ Use diagrams even when not explicitly asked (can earn marks)

☐ For "evaluate" and "discuss," present both sides before concluding

☐ Check calculations; show working for method marks

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